

tion! "Agriculture in India displayed impressive growth, specifically in the area of advanced technology in recent years," adds Saurabh Agarwal.

Artificial intelligence and digital technologies are contributing to the agricultural economy by helping farmers tide over a spate of difficulties. To ensure a rapid adoption rate of these technologies in the farming fraternity across India, significant measures have been taken by the government. "AI possesses the potential to bring about a sea change by bettering food management and decreasing losses along the entire agricultural value chain, i.e., from the farm to the final consumer. Per recent statistics, the penetration of AI injected a surplus of approximately ₹92,651 crore into the agriculture sector," volunteers Saurabh Agarwal.

Aggarwal also adds that the global market size of specifically the AI-driven agricultural sector was valued at 1.1 billion US dollars in 2021, witnessing a CAGR of 25.1% to reach 4.2 billion US dollars by 2028.

Against the backdrop of its agro-based social fabric, India is gradually waking up to embrace the AI-aided ploughing methods. Having seen a green revolution, the nation is now dashing towards artificial intelligence to add another feather in its cap. Fusing established farming practices with new-age technology to gain maximum productivity, Kisan Drones are already being put to use for crop assessments, land records and insecticide spraying, promising to usher in a new dawn of technology in agriculture and farming. Drones with AI, ML and remote-discerning capabilities are becoming increasingly popular.

Artificial intelligence is certainly spreading its tentacles to research and development, and consistently foraying into both private and public sectors. "Feeding of funds, and patronage from government stables and global partners gave rise to agritech startups. However, end-to-end enforcement of AI-led interventions in agricultural network is yet to come in full force," says Navneet Ravikar.

India's Gen-Z is pro AI

Young India comprising millennials and the next-gen represents a pool of brainy talents. This bracket of entrepre-

neurs and software whizkids is floating start-ups and their fledgling businesses are participating in AI-based agriculture by offering sophisticated technologies to solve complex agricultural problems. "Rezo.ai has already collaborated with one of the top agritech companies in imparting farmers with relevant facts and figures. Our AI-triggered bots enable farmers to gain knowledge that can be exploited for better harvests," reports Rashi Gupta.

Farmers across the board in India are trying to warm up to the AI uptake in traditional farming but things take time to happen in phases, admits Aggarwal. "Most farmers are not tech-savvy and therefore, will slowly resort to hi-tech

Bearing in mind that most Indian farmers hail from a humble economic background and many in the community still remain unbanked, will it take ages to make the agrarian society absorb this innovative technology? From a universal perspective, technology has always shown the two sides of the same coin as in cost-intensive products (like smart tractors and devices) and cost-effective solutions (AI bots), opines Gupta. She is optimistic and adds, "The national government is doling out a host of agricultural grants to farmers via concessions on equipment and infrastructural sops, thereby smoothening the swing to the smart agricultural fold. We perceive that the joint efforts from



changes in agricultural practices. However, the new-age progeny that turned to agriculture as its principal occupation is quickly endorsing various technologies and converting the same into profitable businesses," he emphasizes.

The Government

Shedding light on the Government of India's initiatives for automation objectives, Aggarwal confirms that the administration is actively boosting digital agriculture. and says "Recently, the ruling regime released SOPs (standard operating protocols) on drone applications in the agricultural sector. This will enable farmers to use drones for applying agrochemicals, thus declining the risk of susceptibility to harmful chemicals. The Digital Agriculture Mission of 2021-2025 by the Union Government launched in 2021, focuses on accelerating projects based on advanced technologies like AI, blockchain, remote sensing, and the use of drones and robots in agriculture."

the government and startups should engineer this modification and put more money in the farmers' pockets."

Cold Chain: A Strong Link in Supply Chain

"With the swelling world population hitting the 8-billion mark, a quarter of which is dependent on small-holder farmers, threat to food security becomes a prevailing obstacle. The use of artificial intelligence in agriculture impels better resource-efficiency. This is particularly true for climate-smart deeptech-enabled solutions like Ecotron and Ecofrost," states Vivek Pandey, co-founder and CTO of Ecozen Solutions, a Pune-based climate-smart deeptech startup that develops solutions and core technology stacks to deliver a sustainable future, including motor controls, IoT and energy storage. Allocating these technology stacks to the agricultural sector, the company revolutionized cold chains (Ecofrost) and the irrigation industry (Ecotron), thus enormously